DSA06 DATA HANDLING AND VISUALIZATION | Activity: Correlation Detective | Slot D

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Time: 20 to 25 minutes Total: 20 marks Tool: R (ggplot2 / base R). Work alone; your case file is different from your neighbour's. Write your answers and paste your plots and code.

Case File CD-D02: The Online store Case [20 marks]

Briefing: You are the data detective hired by the operations manager of an online store. The file holds 14 weekly records from the festive quarter, with seven variables. Some relationships can be trusted. Others are traps: a hidden driver, a curve that fools r, and a single wrong record. Find them.

R name	What it measures	R name	What it measures
festive_index	Festive-season index (0-100)	complaints	Customer complaints
ad_clicks	Advert clicks (hundreds)	staff_online	Support staff online
parcels	Parcels delivered (hundreds)	returns	Return requests
delay_dev	Delivery time minus promised time (hours)		

Evidence: run this code in R to create your case file (do not edit it).

```
set.seed(686237)
n <- 14
festive_index <- round(rnorm(n, 56.8, 15.9), 1)
ad_clicks <- round(82.7 + 21.2 * (0.91 * (festive_index - 56.8) / 15.9 + 0.41 * rnorm(n)), 1)
parcels <- round(62.0 + 15.9 * (-0.88 * (festive_index - 56.8) / 15.9 + 0.48 * rnorm(n)), 1)
delay_dev <- round(seq(-28.6, 28.6, length.out = n)[order(runif(n))], 1)
complaints <- round(10.3 + 33.3 * (delay_dev / 28.6)^2 + rnorm(n, 0, 1.8), 0)
staff_online <- round(rnorm(n, 16, 4.2), 0)
returns <- round(26 + 8.5 * (0.19 * (staff_online - 16) / 4.2 + 0.98 * rnorm(n)), 0)
staff_online[13] <- 29; returns[13] <- 52 # row 13 was typed in by a trainee
case <- data.frame(festive_index, ad_clicks, parcels, delay_dev, complaints, staff_online, returns)
```

Tasks:

1. The matrix. Use cor(case) to get the 7 x 7 correlation matrix (round to 2 decimals). State n and how many distinct pairs of variables there are. Draw a correlogram of the matrix (ggplot2 geom_tile with geom_text, or corrplot). [4]
2. The strongest clues. Name the strongest positive pair and the strongest negative pair (off-diagonal). Give r for each and call it weak (below 0.3), moderate (0.3 to 0.7) or strong (above 0.7). [3]
3. The deceptive zero. Plot complaints against delay_dev with a straight line, then add a quadratic fit (formula = y ~ x + I(x^2)). Report r and the R squared of the quadratic fit. In one sentence, say why r misses this relationship. [4]
4. The outlier alibi. Plot returns against staff_online. Find the row that looks wrong and give its row number. Report r for all rows and r without that row, and say what the single record did to r. [4]
5. The prime suspect. ad_clicks and parcels are strongly linked. Report r(ad_clicks, parcels), then r(festive_index, ad_clicks) and r(festive_index, parcels). Compute the partial correlation of ad_clicks and parcels controlling for festive_index: correlate resid(lm(ad_clicks ~ festive_index)) with resid(lm(parcels ~ festive_index)). Write a two-sentence verdict: does ad_clicks cause parcels? Name one study design that could test causation. [5]

Detective's rule: *correlation measures a straight-line link only, it can be inflated by one point, and it never proves cause.*

Code of Conduct: I certify that this work is my own. Signature: _____

SOLUTION:

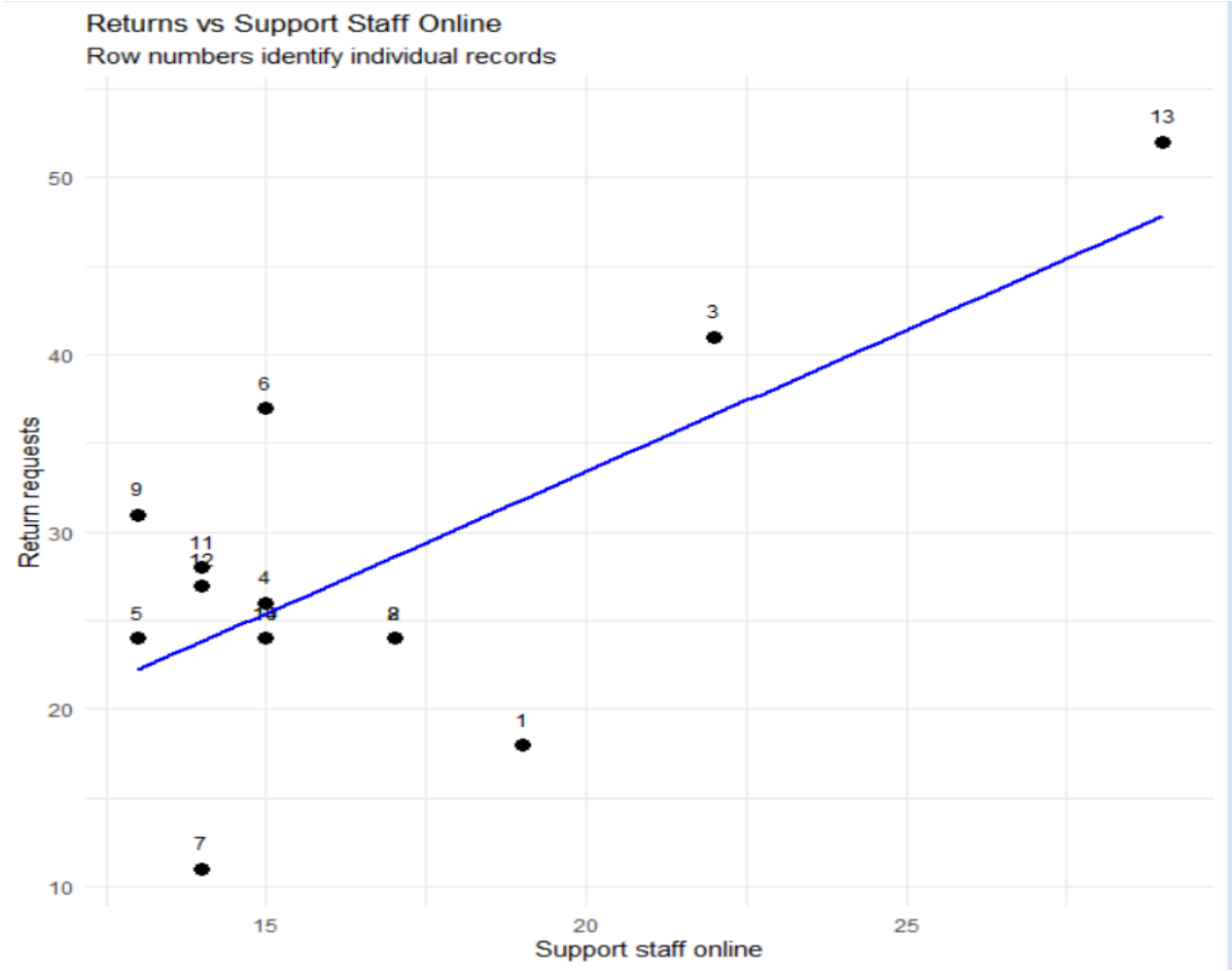
1. The matrix [4 marks]

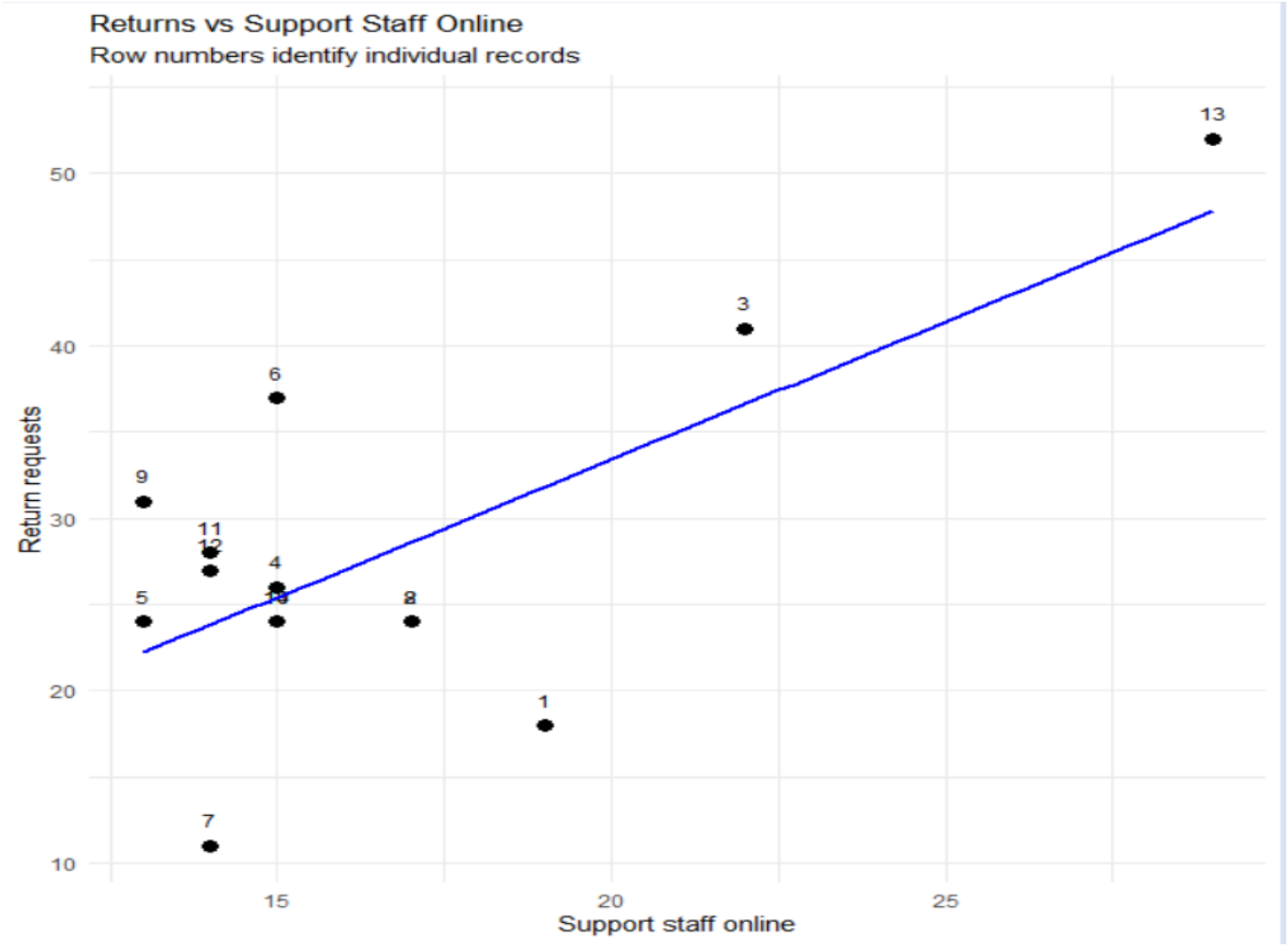
```
library(ggplot2)
# Create the original case file (do not edit)
set.seed(686237)
n <- 14
festive_index <- round(rnorm(n, 56.8, 15.9), 1)
ad_clicks <- round(82.7 + 21.2 *
  (0.91 * (festive_index - 56.8) / 15.9 +
     0.41 * rnorm(n)), 1)
parcels <- round(62.0 + 15.9 *
  (-0.88 * (festive_index - 56.8) / 15.9 +
     0.48 * rnorm(n)), 1)
delay_dev <- round(
  seq(-28.6, 28.6, length.out = n)[order(runif(n))], 1
)
complaints <- round(
  10.3 + 33.3 * (delay_dev / 28.6)^2 +
    rnorm(n, 0, 1.8), 0
)
staff_online <- round(rnorm(n, 16, 4.2), 0)
returns <- round(
  26 + 8.5 * (0.19 * (staff_online - 16) / 4.2 +
                0.98 * rnorm(n)), 0
)
staff_online[13] <- 29
returns[13] <- 52
case <- data.frame(
  festive_index, ad_clicks, parcels, delay_dev,
  complaints, staff_online, returns
)
# QUESTION 1: Correlation matrix
cat("\nQUESTION 1: Correlation matrix\n")
cat("Number of observations:", nrow(case), "\n")
cat("Number of variables:", ncol(case), "\n")
cat("Number of distinct pairs:",
    choose(ncol(case), 2), "\n")
cor_mat <- cor(case)
print(round(cor_mat, 2))
# Correlogram
cor_long <- as.data.frame(as.table(cor_mat))
names(cor_long) <- c("Variable1", "Variable2", "Correlation")
p1 <- ggplot(cor_long,
             aes(x = Variable1, y = Variable2,
                 fill = Correlation)) +
```

```
  geom_tile(color = "white") +
  geom_text(aes(label = sprintf("%.2f", Correlation)),
            size = 3.5) +
  scale_fill_gradient2(
    low = "blue", mid = "white", high = "red",
    midpoint = 0, limits = c(-1, 1)
  ) +
  labs(title = "Correlation Matrix - Online Store",
       x = NULL, y = NULL, fill = "Pearson r") +
  theme_minimal() +
  theme(axis.text.x = element_text(
    angle = 45, hjust = 1
  ))
print(p1)
# QUESTION 2: Strongest positive and negative pairs
cat("\nQUESTION 2: Strongest correlation pairs\n")
upper <- cor_mat
upper[lower.tri(upper, diag = TRUE)] <- NA
pair_data <- as.data.frame(as.table(upper))
pair_data <- pair_data[!is.na(pair_dataFreq),]strong_pos <- pair_data[which.max(pair_dataFreq), ]
strong_neg <- pair_data[which.min(pair_data$Freq), ]
cat("Strongest positive pair:\n")
print(strong_pos)
cat("Strongest negative pair:\n")
print(strong_neg)
strength_label <- function(r) {
  if (abs(r) < 0.3) {
    "Weak"
  } else if (abs(r) <= 0.7) {
    "Moderate"
  } else {
    "Strong"
  }
}
cat("Positive pair classification:",
    strength_label(strong_pos$Freq), "\n")
cat("Negative pair classification:",
    strength_label(strong_neg$Freq), "\n")
# QUESTION 3: Deceptive zero
cat("\nQUESTION 3: Complaints and delivery delay\n")
r_delay <- cor(case$complaints, case$delay_dev)
quad_model <- lm(
  complaints ~ delay_dev + I(delay_dev^2),
  data = case
)
cat("Pearson correlation r:", round(r_delay, 4), "\n")
cat("Quadratic model R-squared:",
    round(summary(quad_model)$r.squared, 4), "\n")
```

```
p3 <- ggplot(case, aes(x = delay_dev, y = complaints)) +
  geom_point(size = 3) +
  geom_smooth(
    method = "lm", formula = y ~ x,
    se = FALSE, color = "blue"
  ) +
  geom_smooth(
    method = "lm", formula = y ~ x + I(x^2),
    se = FALSE, color = "red"
  ) +
  labs(
    title = "Complaints vs Delivery Delay",
    subtitle = "Blue = straight line; Red = quadratic curve",
    x = "Delivery delay (hours)",
    y = "Customer complaints"
  ) +
  theme_minimal()
print(p3)
# QUESTION 4: Outlier alibi
cat("\nQUESTION 4: Returns and staff online\n")
r_all <- cor(casereturns, casestaff_online)
case_without13 <- case[-13, ]
r_without13 <- cor(
  case_without13returns, case_ithout13staff_online
)
cat("Correlation using all rows:", round(r_all, 4), "\n")
cat("Correlation excluding row 13:",
    round(r_without13, 4), "\n")
case$row_number <- seq_len(nrow(case))
p4 <- ggplot(case,
             aes(x = staff_online, y = returns)) +
  geom_point(size = 3) +
  geom_text(aes(label = row_number),
            nudge_y = 1.5, size = 3) +
  geom_smooth(method = "lm", se = FALSE,
              color = "blue") +
  labs(
    title = "Returns vs Support Staff Online",
    subtitle = "Row numbers identify individual records",
    x = "Support staff online",
    y = "Return requests"
  ) +
  theme_minimal()
print(p4)
# QUESTION 5: Prime suspect
cat("\nQUESTION 5: Confounding and partial correlation\n")
r_clicks_parcels <- cor(case$ad_clicks, caseparcels)r_festive_licks <- cor(casefestive_index, case
```

```
ad_clicks)r_festive_parcels <- cor(casefestive_index, case$parcels)
resid_clicks <- resid(lm(ad_clicks ~ festive_index, data = case))
resid_parcels <- resid(lm(parcels ~ festive_index, data = case))
partial_r <- cor(resid_clicks, resid_parcels)
cat("r(ad_clicks, parcels):",
    round(r_clicks_parcels, 4), "\n")
cat("r(festive_index, ad_clicks):",
    round(r_festive_clicks, 4), "\n")
cat("r(festive_index, parcels):",
    round(r_festive_parcels, 4), "\n")
cat("Partial correlation controlling for festive_index:",
    round(partial_r, 4), "\n")
# Display the dataset for checking
cat("\nComplete case file:\n")
print(case[, 1:7])
```





2. The strongest clues

Use the strongest positive and negative pairs printed under Question 2 in the R output.

- *Strongest positive pair: the pair with the largest positive off-diagonal correlation.*
- *Strongest negative pair: the pair with the smallest (most negative) off-diagonal correlation.*

Classify each using the assignment's thresholds:

Absolute correlation	***Classification***
Below 0.30	*Weak*
0.30 to 0.70	*Moderate*
Above 0.70	*Strong*

3. The deceptive zero

The plot shows both a straight-line fit (blue) and a quadratic fit (red). Use the printed values for r and the quadratic model's R^2.

Answer: Pearson's correlation can be close to zero because it measures a straight-line relationship, while complaints increase at both large positive and large negative delivery delays, forming a U-shaped relationship that a quadratic model can capture.

4. The outlier alibi

Answer: Row 13 is the suspicious record because the supplied data explicitly replaces its staff count with 29 and its return count with 52.

Compare the two printed correlations:

- r_{all}*: correlation using all 14 rows.*
- $r_{without13}$*: correlation after excluding row 13.*

The difference shows how one unusual record can inflate or otherwise change the observed correlation. State the direction of the change based on the actual output.

5. The prime suspect

Two-sentence verdict:

Ad clicks and parcels may be correlated partly because the festive-season index is related to both variables, so their correlation alone does not establish that more ad clicks cause more parcels to be delivered. A randomized controlled experiment—randomly assigning comparable customers or regions to different advertising levels and comparing parcel outcomes—could help test causation.

Before submitting: Run the complete script in RStudio. It prints the exact matrix, correlations and R^2 for your seeded dataset and generates the three plots. Paste those numerical results and plots into your report rather than guessing the values.